








Time Series Analysis

Seminar 14: Multivariate GARCH models



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **CCC**: Constant Conditional Correlation (model)
- **DCC**: Dynamic Conditional Correlation
- **EODHD**: EOD Historical Data (market-data provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **LR**: Likelihood Ratio (test)
- **OLS**: Ordinary Least Squares
- **US**: United States
- **VEC**: Vector (half-vectorised) GARCH model

Today's Question and Seminar Structure

- The question of the seminar
  - **how do correlations move from day to day, and what does that change for the risk of a portfolio and for a hedge?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C1
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Chapter 14 is a self-study chapter: read it before the seminar; the recap gives the formulas used today
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	one step of the DCC recursion on 17 March 2020; one step of a diagonal BEKK	Solved, Proposed	A1
A3, A4	portfolio variance and VaR (value at risk) under CCC and DCC; the hedge ratio of the BET	Solved, Proposed	A3
A5	the half-life of a correlation shock	Solved	A5
B1, B2	stocks and US Treasuries; a VaR backtest of a BET and DAX portfolio	Solved, Proposed	B1
C1	what is wrong in an AI answer?	Proposed	A1–B2

- Types: **A** calculations on paper (with numbers from real data); **B** real data, in the notebook; **C** an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
S&P 500, DAX	EODHD (GSPC.INDX, GDAXI.INDX)	daily, close	2000–2026
BET	EODHD (BET)	daily, close	2005–2026
TLT (US Treasuries over 20 years)	EODHD (TLT.US)	daily, adjusted close	2003–2026

- Daily log returns in % on the days on which both markets trade (`common_returns`); data up to 18 September 2026
- In the notebook: `step1` (GARCH(1,1) per series), `dcc_fit`, `dcc_path`, `kupiec`, `christoffersen` (the functions of Chapter 14); no account or key is needed

Recap (1/2): CCC, DCC and BEKK

- **Conditional covariance** of N returns: $\varepsilon_t = r_t - \mu$, $\mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-\mathbf{1}}) = H_t$

$$H_t = D_t R_t D_t, \qquad D_t = \mathrm{diag}(\sigma_{\mathbf{1},t}, \ldots, \sigma_{N,t}), \qquad z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$$

  - $\sigma^{\mathbf{2}}_{i,t}$: a GARCH(1,1) per series (step 1); CCC: $R_t = \bar{R}$ constant Bollerslev (1990)

- **DCC** Engle (2002): step 2, with $\bar{Q}$ the sample correlation of z_t

$$Q_t = (1 - a - b)\,\bar{Q} + a\, z_{t-\mathbf{1}} z_{t-\mathbf{1}}^\top + b\, Q_{t-\mathbf{1}}, \qquad \rho_{\mathbf{12},t} = \frac{q_{\mathbf{12},t}}{\sqrt{q_{\mathbf{11},t}\, q_{\mathbf{22},t}}}$$

  - $a, b \geq 0$, $a + b < 1$; $a = b = 0$ gives CCC (likelihood-ratio test, 2 degrees of freedom); half-life of a shock $\ln 0.5 / \ln(a + b)$

- **Diagonal BEKK** Engle and Kroner (1995): $H_t = CC^\top + A\varepsilon_{t-\mathbf{1}}\varepsilon_{t-\mathbf{1}}^\top A + BH_{t-\mathbf{1}}B$, A, B diagonal

$$h_{ij,t} = c_{ij} + a_{ii}a_{jj}\,\varepsilon_{i,t-\mathbf{1}}\varepsilon_{j,t-\mathbf{1}} + b_{ii}b_{jj}\,h_{ij,t-\mathbf{1}}$$

  - c_{ij}: elements of $CC^\top$; H_t is positive definite by construction

Recap (2/2): portfolio risk, hedging and backtests

- **Portfolio** with weights w: mean $\mu_p = w^\top \mu$ and

$$\sigma^2_{p,t} = w^\top H_t w = w_1^2 h_{11,t} + w_2^2 h_{22,t} + 2 w_1 w_2 h_{12,t}, \qquad \mathrm{VaR}_{\alpha,t} = -(\mu_p + z_\alpha \sigma_{p,t})$$

  - VaR (value at risk) 1%: $z_{0.01} = -2.326$ (Normal); a loss larger than the VaR is a **violation**
- **Minimum-variance hedge** of a spot position s with an instrument f Kroner and Sultan (1993)

$$h_t = \frac{\mathrm{Cov}_t(r_s, r_f)}{\mathrm{Var}_t(r_f)} = \rho_t \frac{\sigma_{s,t}}{\sigma_{f,t}}, \qquad \text{effectiveness} = 1 - \frac{\mathrm{Var}(r_s - h\, r_f)}{\mathrm{Var}(r_s)} = \rho^2 \text{ (constant } \rho)$$

- **Backtest** over n days with x violations, $\hat{p} = x/n$ Kupiec (1995)

$$\mathrm{LR}_{uc} = -2 \ln \frac{(1-\alpha)^{n-x}\alpha^x}{(1-\hat{p})^{n-x}\hat{p}^x} \sim \chi^2(1), \qquad \text{reject at 5\% if } \mathrm{LR}_{uc} > 3.84$$

  - Christoffersen: are violations independent (a violation today does not make one tomorrow more likely)? Christoffersen (1998)

A1: one step of the DCC recursion [Solved]

Task

- S&P 500 and DAX, DCC estimated on 2000–2026: $\hat{a} = 0.0152$, $\hat{b} = 0.9766$, $\bar{q}_{12} = 0.5748$. On 16 March 2020: returns -12.77% and -5.45%, $z = (-2.274,\ -1.182)$; $q_{11} = 1.5509$, $q_{22} = 1.7093$, $q_{12} = 1.1471$.
- 1. Compute q_{11}, q_{22}, q_{12} for 17 March 2020, term by term.
- 2. Compute the correlation of 16 March 2020 and of 17 March 2020.
- 3. Why does a fall of 12% move the correlation so little?
- Report: three elements of Q_t, two correlations, one sentence.

Solution

- 1. $1 - a - b = 0.0082$; $q_{12} = 0.0047 + 0.0409 + 1.1202 = 1.1659$
- $q_{11} = 0.0082 + 0.0788 + 1.5147 = 1.6016$; $q_{22} = 0.0082 + 0.0213 + 1.6693 = 1.6987$
- 2. $\rho = 1.1471/\sqrt{1.5509 \cdot 1.7093} = 0.7045$, then $1.1659/\sqrt{1.6016 \cdot 1.6987} = 0.7068$
- 3. b is close to 1 and a is small, and z is the return divided by an already high GARCH volatility: the residuals are moderate and the old Q_{t-1} dominates

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Back Exercise list

A2: one step of a diagonal BEKK [Proposed]

Solution discussed in class

Task

- $c_{11} = 0.05$, $c_{12} = 0.01$, $c_{22} = 0.04$; $a_{11} = 0.25$, $a_{22} = 0.30$; $b_{11} = 0.95$, $b_{22} = 0.93$; $\varepsilon_{t-1} = (1.5,\ -1.0)$; $h_{11} = 1.2$, $h_{12} = 0.3$, $h_{22} = 0.9$ at $t-1$. Model: A1.
- 1. Compute $h_{11,t}$, $h_{22,t}$, $h_{12,t}$ term by term.
- 2. Compute the correlation at $t-1$ and at t. Why does it fall?
- 3. Check that H_t is positive definite; give the persistences $a_{ii}^2 + b_{ii}^2$ and $a_{11}a_{22} + b_{11}b_{22}$.
- Report: three elements, two correlations, one determinant, three persistences.

A3: the data [Solved]

Model	σ_1	σ_2	ρ_{12}	h_{12}	Note
DCC	6.888	4.727	0.707	?	GARCH volatilities, DCC correlation (A1)
CCC	6.888	4.727	0.575	?	GARCH volatilities, $\bar{q}_{12}$
Static	1.221	1.429	0.599	?	sample covariance 2000–2026

- S&P 500 (1) and DAX (2) on 17 March 2020; daily volatilities in % (annualised: 109% and 75%); portfolio mean $\mu_p = 0.065\%$, weights $w = (0.5,\ 0.5)$, $z_{0.01} = -2.326$

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A3: portfolio variance and VaR under CCC and DCC [Solved]

Task

- The table of the previous slide.
- 1. Compute $h_{12} = \rho_{12}\sigma_1\sigma_2$ for each model.
- 2. Compute $\sigma_p^2 = w^\top H w$, σ_p and the VaR 1% for each model.
- 3. Which part of the VaR comes from the correlation, which from the volatilities?
- Report: three covariances, three VaRs, one sentence.

Solution

- 1. DCC $h_{12} = 23.016$; CCC 18.718; static 1.045
- 2. DCC: $\sigma_p^2 = 11.862 + 5.587 + 11.508 = 28.956$, $\sigma_p = 5.381$, VaR $= 12.45\%$
- CCC: $\sigma_p = 5.178$, VaR 11.98%; static: $\sigma_p = 1.186$, VaR 2.69%
- 3. The volatilities do most of the work (static against CCC); the higher crisis correlation adds the rest (CCC against DCC). The return of that day was +4.02%: no violation

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A4: the hedge ratio of the BET with the DAX [Proposed]

Solution discussed in class

Task

- DCC of the BET and the DAX estimated on 2005–2014 (Chapter 14). On 18 September 2026: $\rho_t = 0.253$, $\sigma_{BET,t} = 1.735\%$, $\sigma_{DAX,t} = 0.832\%$ (daily). Model: A3.
- 1. Compute the hedge ratio h_t and the DAX position for 100 lei of BET.
- 2. Compute the effectiveness ρ_t^2 and the volatility of the hedged position $\sigma_s\sqrt{1-\rho^2}$.
- 3. Compare with the static OLS ratio (2005–2014) and with the out-of-sample effectiveness of the lecture.
- Report: two ratios, one position, two effectiveness figures, one sentence.

A5: the half-life of a correlation shock [Solved]

Task

- Estimated $a + b$: S&P 500 and DAX 0.9918 (A1); the five-asset panel of the lecture 0.9870; S&P 500 and TLT 0.9846 (B1).
- 1. Compute the half-life $\ln 0.5 / \ln(a + b)$ and the days until 90% of a shock is gone, $\ln 0.1 / \ln(a + b)$.
- 2. With the approximation $\rho_{t+k} \approx \bar{\rho} + (a + b)^k (\rho_t - \bar{\rho})$, forecast the S&P 500 and DAX correlation 10 and 50 days after 17 March 2020.
- 3. What is the half-life under CCC?
- Report: six durations, two forecasts, one sentence.

Solution

- 1. S&P 500 and DAX: 84.6 and 281 days; panel: 53.0 and 176; stocks and bonds: 44.6 and 148
- 2. $\rho_t - \bar{\rho} = 0.707 - 0.575 = 0.132$; $k = 10$: 0.696; $k = 50$: 0.662
- 3. Under CCC there is nothing to decay ($a + b = 0$, the correlation never moves)
- a correlation raised by a crisis stays high for months: diversification returns slowly

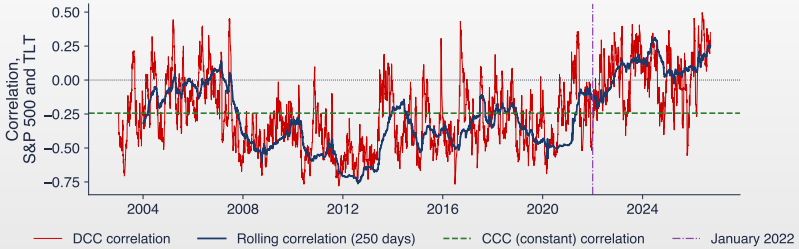
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Back Exercise list

B1: stocks and US Treasuries [Solved]

- **Question**: do long Treasuries still hedge stocks, or did the stock-bond correlation change sign after 2022?
- **Inputs**: daily S&P 500 and TLT log returns on common trading days, 5 933 days from January 2003 to September 2026
- **Tasks**
  1. Fit a GARCH(1,1) to each series and the DCC by two-step maximum likelihood (`step1`, `dcc_fit`).
  2. Test CCC against DCC with the likelihood ratio; compute the half-life of a correlation shock.
  3. Plot the DCC and the 250-day rolling correlation; report the mean correlation before and after January 2022.
  4. Interpretation: what does the change mean for a 60/40 stock-bond portfolio?
- **Report**: four parameters with standard errors, one test, one half-life, two means, one sentence
- Notebook: section B1

B1: solution [Solved]



- $\hat{a} = 0.0530$ (0.0060), $\hat{b} = 0.9315$ (0.0086), $a + b = 0.9846$, half-life 44.6 days; LR $= 494.4 > 5.99$: constant correlation rejected; $\bar{q}_{12} = -0.244$
- Mean DCC correlation -0.304 before 2022 (positive on 12% of the days), 0.040 since January 2022 (positive on 59%); minimum -0.780 (2 November 2011), maximum 0.497 (21 May 2026)
- Interpretation: for twenty years bonds rose when stocks fell; since the inflation shock of 2022 they often fall together, so a 60/40 portfolio is less diversified than its history suggests

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B2: a VaR backtest of a BET and DAX portfolio [Proposed]

- **Question**: does a DCC VaR 1% give the right number of violations for a Romanian investor who holds half BET and half DAX?
- **Inputs**: daily BET and DAX log returns, parameters estimated on 2005–2014, VaR computed out of sample on 2015–2026 ($n = 2\,893$ days); model: A3 and the VaR of Chapter 14
- **Tasks**
  1. Compute the one-day VaR 1% of the 50/50 portfolio with DCC, with CCC and with the static covariance of 2005–2014.
  2. Count the violations and compare with the expected number $0.01n$.
  3. Apply the Kupiec and Christoffersen tests (`kupiec`, `christoffersen`).
  4. Interpretation: why can the static VaR pass the count and still be a poor risk measure?
- **Report**: three counts, three tests, one sentence
- Notebook: section B2

C1: audit an AI answer [Proposed]

- A student asked an AI assistant about multivariate GARCH models. The answer:
- `(a) A full BEKK(1,1) for 10 assets has about as many parameters as a CCC model.`
- `(b) For the S&P 500 and the DAX, a + b = 0.992, so a correlation shock is gone within a week.`
- `(c) CCC is DCC with a = b = 0; the LR statistic of 131.8 rejects constant correlation.`
- `(d) The VaR of a portfolio is the weighted sum of the VaRs of its assets.`
- `(e) 65 violations of a VaR 1% in 2893 days is close to 1%, so the model is well calibrated.`
- `(f) The daily correlation of the S&P 500 and the BET is the right input for a weekly risk report.`
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (Parts A and B, notebook section C1).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- DCC = GARCH volatilities + a correlation recursion with two parameters; CCC is the case $a = b = 0$
- $a + b$ close to 1: correlation shocks last months; in a crisis volatilities and correlations rise together
- Portfolio risk is $\sqrt{w^\top H_t w}$, not a sum of risks; the hedge ratio is $\rho_t \sigma_{s,t} / \sigma_{f,t}$
- The stock-bond correlation changed sign after 2022: a constant correlation hides it
- A VaR model is judged by its violations: their number and their clustering

After the seminar

- Chapter 14 (self-study) develops each topic of today
  - the models and their parameters, the two-step estimation, the five-asset panel, VaR with filtered historical simulation, hedging
- Try the [Proposed] tasks in the notebook
- A project idea: a DCC for the BET, the EUR/RON reference rate and Romanian 10-year yields
- Reading: Engle (2002); Bollerslev (1990); Engle and Kroner (1995); Kupiec (1995)

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